Systematic review

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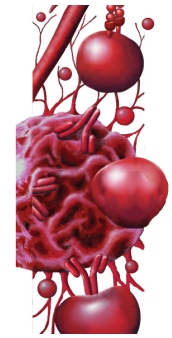

Immunotherapy in advanced melanoma: a network meta-analysis

Aim: The aim of this study was to compare the effects of various immunotherapeutic agents and chemotherapy for unresected or metastatic melanomas. **Methods:** We performed a network meta-analysis using a Bayesian statistical model to compare objective response rate (ORR) of various immunotherapies from 12 randomized controlled studies. **Results:** The estimated ORRs of immunotherapy and chemotherapy were 0.224 and 0.108, respectively. The ORRs of immunotherapy in untreated and pretreated patients were 0.279 and 0.176, respectively. In network meta-analysis, the odds ratios for ORR of nivolumab (1 mg/kg)/ipilmumab (3 mg/kg), pembrolizumab 10 mg/kg and nivolumab 3 mg/kg were 8.54, 5.39 and 4.35, respectively, compared with chemotherapy alone. **Conclusion:** Our data showed that various immunotherapies had higher ORRs rather than chemotherapy alone.

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Keywords: immunotherapy • malignant melanoma • network meta-analysis • objective response rate • traditional meta-analysis

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The incidence of melanoma continues to rise during the past 40 years despite most cancer types are stable or declining [1]. Advanced melanoma, including metastatic and unresectable cases, accounts for 4% of all melanoma at diagnosis [2]. The median overall survival is 6.2 months and the 1-year survival rate is 25.5% for stage IV melanoma patients [3]. The 5-year survival for patients with metastatic melanoma is less than 16% [4]. Melanoma is treated with a combination of surgery, traditional cytotoxic chemotherapy, targeted therapies and immune-based therapies. Due to progress in the understanding of melanoma biology, breakthroughs in both molecular-targeted therapies and immunotherapies have revolutionized the standard of care for patients with advanced melanoma [5]. *BRAF* mutation that is found in 40–60% of cutaneous melanomas is one of the targets of therapeutic strategies [6–8]. Immunotherapy with checkpoint inhibitors involves the augmentation of cell-mediated immunity and anticancer immune response by blocking key checkpoint molecules [9,10].

To date, the most clinically advanced checkpoint inhibitors are blocking monoclonal antibodies that target CTLA-4 and PD-1 pathways. Ipilimumab (antibody against CTLA-4), pembrolizumab and nivolumab (antibodies against PD-1) are the three agents approved by the US FDA for treating advanced melanoma [11]. These immunotherapeutic agents have shown survival gain and definitive objective response rate [ORR] in randomized controlled trials (RCTs) [12–23]. In addition, they have less adverse effects than chemotherapy alone [17,22]. In general, pembrolizumab and nivolumab can be used before ipilimumab. A recent pooled analysis of overall survival has provided evidence supporting the durability of long-term survival in a proportion of ipilimumab-treated

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patients with advanced melanoma [24]. However, direct comparisons between various immunotherapeutic modalities have not been fully investigated. In addition, the therapeutic effects of immunotherapies in various populations such as patients previously treated by chemotherapy or immunotherapy and *BRAF* wild-type patients remain unclear. Therefore, the objective of this study was to determine the ORRs of various immunotherapies and chemotherapy alone in advanced melanomas through traditional meta-analysis. Using network meta-analysis, comparisons of ORRs between various immunotherapies and chemotherapy alone were conducted. In addition, subgroup analysis based on previous treatment was performed to confirm the therapeutic effects of immunotherapy in advanced melanomas.

Materials & methods

Search strategy for published study

Relevant articles were obtained by searching PubMed and MEDLINE databases through 15 June 2016. These databases were searched using the keywords 'melanoma', 'randomized controlled trial (RCT)' and 'immunotherapy, ipilimumab, nivolumab or pembrolizumab'. The titles and abstracts of all searched articles were screened for exclusion. Review articles were also screened to find additional eligible studies.

Selection criteria

Articles were included if the study was performed on human melanomas and if there was information about the ORR of melanoma patients after immunotherapy. All RCTs that compared at least two different treatment regimens involving immunotherapy or chemotherapy were obtained. Articles were excluded if they were RCTs or if they were not written in English.

Data extraction

Data from all eligible studies were extracted by two independent authors. The included data were extracted from each of the eligible studies [12–23], including the first author's name, year of publication, randomized phase, characteristics and number of enrolled patients, and regimens of immunotherapy and chemotherapy.

Data analyses

To perform traditional meta-analysis, data were analyzed using the Comprehensive Meta-Analysis software package (Biostat, NJ, USA). We investigated the ORR of melanoma patients in various immunotherapy and chemotherapy groups. The ORR defined as complete response or partial response was according to Response Evaluation Criteria in Solid Tumors. Subgroup analysis was performed based on the combination of chemotherapy and the presence of previous treatment. Heterogeneity between studies was checked by the Q and R statistics and expressed as p-values. Additionally, sensitivity analysis was conducted to assess the heterogeneity of eligible studies and the impact of each study on combined effects. To compare the differences between subgroups, the meta-regression test was performed. In the meta-analysis, because eligible studies used various populations, a random-effect model rather than a fixed-effect model was used. To assess publication bias, Begg's funnel plot and Egger's test were used. Results were considered as statistically significant when the p-value was less than 0.05.

For network meta-analysis, the Bayesian consistency model as described in NetMetaXL was used. To provide an interface for conducting a Bayesian network meta-analysis, WinBUGS 1.4.3 (MRC Biostatistics Unit, Cambridge, UK) was used [25]. We specified the number of iterations and burn-ins as 100,000 for assessing convergence of parameter estimates. The evidence of network was first generated. Width of each edge was proportional to the number of RCTs comparing each pair of treatments and the size of each treatment node was proportional to the number of randomized participants. To compare the therapeutic effect, odds ratios (ORs) for ORRs of two treatment modalities were calculated. When the corresponding value of OR was over 1, the therapeutic effect between two therapeutic modalities was better. Values of surface under cumulative ranking probability curve (SUCRA) were calculated through the above program. SUCRA would be 1 when a treatment is certain to be the best or 0 when a treatment is certain to be the worst [25].

Results

Selection & characteristics of studies

In the present study, 105 reports were identified in the database search. Among the searched reports, 36 were excluded due to lack of information or with insufficient information. In addition, 57 reports focused on other diseases (n = 6), nonoriginal articles (n = 34), duplicated articles (n = 10), used animals or cell lines (n = 3) or articles in a language other than English (n = 4) were excluded. Twelve RCTs were ultimately included. A traditional meta-analysis and a network meta-analysis were performed (Figure 1). The characteristics and the evidence of network of eligible studies are shown in Table 1 and Figure 2, respectively [12–23]. Direct comparisons among ipilimumab alone, nivolumab/ipilimumab and chemotherapy alone were not present (Figure 2A). In addition, in untreated patients, there were no direct comparisons among nivolumab/ipilimumab, pembrolizumab, ipilmumab and chemotherapy alone (Figure 2B).

Comparison of ORR from traditional meta-analysis

We first analyzed the ORRs after various treatments through traditional meta-analysis. The estimated ORRs of immunotherapy and chemotherapy were 0.224 (95% confidence interval [CI]: 0.169–0.290) and 0.108 (95% CI: 0.071–0.161), respectively. In subgroup treated immunotherapy, the ORRs of immunotherapy alone and combined immunotherapy and chemotherapy were 0.222 (95% CI: 0.165–0.292) and 0.239 (95% CI: 0.087–0.508), respectively (Table 2). In untreated and pretreated patients, the ORRs of immunotherapy were 0.279 (95% CI: 0.195-0.383) and 0.176 (95% CI: 0.120–0.251), respectively. The ORRs of chemotherapy in untreated and pretreated patients were 0.131 (95% CI: 0.095–0.177) and 0.045 (95% CI: 0.023–0.087), respectively. In subgroup analysis based on previous treatment modality, the ORRs of immunotherapy of patients previously treated with immunotherapy and chemotherapy were 0.239 (95% CI: 0.204–0.277) and 0.051 (95% CI: 0.015–0.159), respectively. In subgroup analysis based on immunotherapeutic agents, the estimated ORR of nivolumab-based immunotherapy was 0.462 (95% CI: 0.370–0.557), higher than that of pembrolizumab- and ipilmumab-based immunotherapies (0.264, 95% CI: 0.212–0.324 and 0.129, 95% CI: 0.089–0.185, respectively).

Network meta-analysis

The ORs for ORR of nivolumab (1 mg/kg)/ipilmumab (3 mg/kg), pembrolizumab 10 mg/kg, pembrolizumab 2 mg/kg and nivolumab 3 mg/kg regimens were 8.54 (95% CI: 3.78–21.70), 5.39 (95% CI: 2.52–12.18), 4.64 (95% CI: 2.00–11.65) and 4.35 (95% CI: 2.33–11.79), respectively, compared with chemotherapy alone (Figure 3A). The ORs for ORRs of ipilimumab 10 mg/kg alone, combination of ililmumab (10 mg/kg) and chemotherapy, and ipilimumab 3 mg/kg alone were 2.93 (95% CI: 0.75–11.79), 2.20 (95% CI: 1.20–4.83) and 1.27 (95% CI: 0.57–2.60), respectively, which were higher than that of chemotherapy alone. However, the ORR of ipilimumab 0.3 mg/kg alone was lower than that of chemotherapy alone. The values of SUCRA were 0.9626 and 0.7923 for nivolumab (1 mg/kg)/ipilmumab (3 mg/kg) and pembrolizumab 10 mg/kg, respectively. In chemotherapy alone, the value of SUCRA was 0.1599. In rank probability test, combined nivolumab (1 mg/kg) and ipilmumab (3 mg/kg) was the best treatment modality. In untreated patients, the ORs of nivolumab (1 mg/kg)/ipilmumab (3 mg/kg) compared with nivolumab 3 mg/kg, pembrolizumab 10 mg/kg and chemotherapy alone were 7.47 (95% CI: 1.09–55.54), 1.90 (95% CI: 0.43–9.72) and 1.99 (95% CI: 0.26–20.15), respectively (Figure 3B). The value of SUCRA of nivolumab (1 mg/kg)/ipilmumab (3 mg/kg) in untreated patients was 0.9200, which was superior to other treatment modalities.

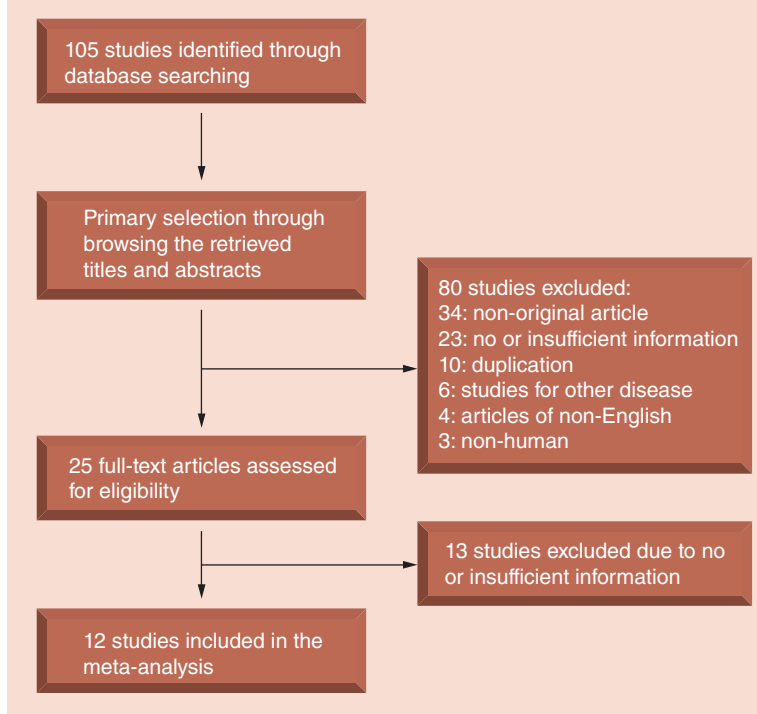

Figure 1. Flow chart for study search and selection methods.

Discussion

Selective BRAF inhibitors and combined BRAF/MEK inhibitors are standard treatment for patients with *BRAF* V600E-mutant advanced melanoma [26]. Recently, various immunotherapeutic agents have been introduced and studied in advanced melanomas. The current study was the first network meta-analysis of published randomized controlled studies to determine the therapeutic effects of various immunotherapy on advanced melanomas.

Currently, various treatment modalities such as chemotherapeutic agents, targeted agent and immunotherapeutic agents, are applied to advanced melanomas. However, direct comparisons between various treatment modalities have not been fully investigated. Dequen *et al.* have reported a direct comparison for overall survival among anti-CTLA-4 antibody, ipililmumab and traditional chemotherapeutic agent for pretreated unresectable melanomas [2]. However, there was no information for ORR in that network meta-analysis [2]. Mai *et al.* have reported the results of a network meta-analysis to determine the therapeutic effect of targeted therapy using a combination of BRAF and MEK inhibitors [8]. The ORR of tar-

Table 1. Main characteristics of the eligible studies.

Study (year)	Phase	Characteristics	Arm	Regimen	Number of patients	
					Enrolled	CR & PR
Hamid (2011)	2	Pre- and un-treated pts	Arm 1	Ipilimumab 10 mg/kg, Q3W	42	5
			Arm 2	Ipilimumab 3 mg/kg, Q3W	40	3
Hersh (2011)	2	CTx-naive pts	Arm 1	Ipilimumab 3 mg/kg, Q4W + dacarbazine 250 mg/m^2, Q3W	35	5
			Arm 2	Ipilimumab 3 mg/kg, Q4W	37	2
Larkin (2015)	3	Untreated pts.	Arm 1	Nivolumab 1 mg/kg + ipilimumab 3 mg/kg, Q3W	314	181
			Arm 2	Nivolumab 3 mg/kg, Q2W	316	138
			Arm 3	Ipilimumab 3 mg/kg, Q2W	315	60
Maio (2015)	3	Untreated pts	Arm 1	Ipilimumab 10 mg/kg + dacarbazine 850 mg/m^2, Q3W	40	20
			Arm 2	Dacarbazine 850 mg/m^2, Q3W	35	7
Postow (2015)	1	Untreated pts	Arm 1	Nivolumab 1 mg/kg + ipilimumab 3 mg/kg, Q3W	95	56
			Arm 2	Ipilimumab 3 mg/kg, Q3W	47	5
Ribas (2015)	2	Ipilimumab refractory pts	Arm 1	Pembrolizumab 10 mg/kg, Q3W	181	46
			Arm 2	Pembrolizumab 2 mg/kg, Q3W	180	36
			Arm 3	CTx (paclitaxel + carboplatin, paclitaxel, carboplatin, dacarbazine, oral temozolomide)	179	8
Robert (2015) (a)	3	Untreated pts	Arm 1	Pembrolizumab 10 mg/kg, Q2W	279	94
			Arm 2	Pembrolizumab 10 mg/kg, Q3W	277	91
			Arm 3	Ipilimumab 3 mg/kg, Q3W	278	33
Robert (2015) (b)	3	Untreated pts	Arm 1	Nivolumab 3 mg/kg, Q2W	210	84
			Arm 2	Dacarbazine 1000 mg/m^2, Q3W	208	29
Robert (2014)	1	Pretreated pts	Arm 1	Pembrolizumab 10 mg/kg, Q3W	76	20
			Arm 2	Pembrolizumab 2 mg/kg, Q3W	81	21
Robert (2011)	3	Untreated pts	Arm 1	Ipilimumab 10 mg/kg + dacarbazine 850 mg/m^2, Q3W	250	38
			Arm 2	Dacarbazine 850 mg/m^2, Q3W	252	26
Weber (2015)	3	Pre- and un-treated pts	Arm 1	Nivolumab 3 mg/kg, Q2W	120	38
			Arm 2	Dacarbazine 1000 mg/m^2, Q3W or carboplatin AUC 6 + paclitaxel 175 mg/m^2, Q3W	47	5
Wolchok (2010)	2	Pretreated pts	Arm 1	Ipilimumab 10 mg/kg, Q3W	72	8
			Arm 2	Ipilimumab 3 mg/kg, Q3W	72	3
			Arm 3	Ipilimumab 0.3 mg/kg, Q3W	73	0

CR & PR: Complete and partial response; CTx: Chemotherapy; Pts: Patients; Q2W: Per every 2 week; Q3W: Per every 3 week; Q4W: Per every 4 week.

geted therapy was found to be higher than that of traditional chemotherapy alone in their network meta-analysis [8]. Comparisons between immunotherapies have only been reported for nivolumab and ipilimumab in previous RCTs [12–23]. No direct comparisons have been made between combined nivolumab/ipilimumab and chemotherapy alone, ipilimumab, and pembrolizumab in previous RCTs (Figure 2). The network meta-analysis performed in this study will be useful for interpreting the effects of various immunotherapies or between immunotherapy and chemotherapy.

In previous studies, treatments using anti-PD-1 and anti-CTLA-4 antibodies showed 30–40% [21,27] and 10–15% of ORR, respectively [19,22]. The complementary and nonredundant mechanisms of PD-1 and CTLA-4 provide a rationale for combination therapy. Earlier animal studies have demonstrated that both inhibitors can act in synergy [28]. Strikingly, Chapman *et al.* have recently reported a case of rapid eradication of a large melanoma mass after a single treatment with a combination of nivolumab and ipilimumab [29]. Our current network meta-analysis revealed that combined nivolumab (1 mg/kg)/ipilimumab (3 mg/kg) had the highest ORs and SUCRA value compared with other treatment modalities. Our findings further support the NCCN guidelines that anti-PD-1 monotherapy or in combination with ipilimumab are the first choices for melanoma patients without *BRAF* mutation.

Previous studies have shown that the expression of programmed cell death ligand 1 (PD-L1) is associated with high ORRs in patients treated with nivolumab alone [14,22,30–31]. However, the correlation between PD-L1 expression and the therapeutic effect of immunotherapy has not been fully elucidated. In combination of nivolumab/ipilimumab, the correlation between PD-L1 expression and therapeutic effect is controversial [14,16]. In addition, using anti-PD-1 antibody as indication for immunotherapy is unclear, although these treatment modalities have been applied in melanomas with progression after ipilimumab therapy or *BRAF* mutated melanomas after BRAF inhibition therapy. In our unpublished data, the rate of PD-L1 IHC expression in melanomas was 0.537 (95% CI: 0.349–0.714). There was no significant correlation between PD-L1 expression and worse survival rate. These results suggest that negative expression of PD-L1 should not be exclusion criteria for immunotherapy.

In our traditional meta-analysis, the ORRs of immunotherapy and chemotherapy alone in untreated patients were 0.279 (95% CI: 0.195–0.383) and 0.131 (95% CI: 0.095–0.177), respectively. There was no significant difference between the two groups (p = 0.091). In untreated patients, there have been no direct comparisons among chemotherapy alone, pembrolizumab 10 mg/kg, nivolumab/ipilimumab and ipilimumab 3 mg/kg. In the current network meta-analysis for untreated patients, the ORs for ORR in nivolumab/ipilimumab, nivolumab, pembrolizumab 10 mg/kg and ipilimumab/CTx were 7.47 (95% CI: 1.09–55.54), 3.92 (95% CI: 0.82–17.78), 3.73 (95% CI: 0.27–41.29) and 2.29 (95% CI: 0.75–8.71), respectively, compared with chemotherapy alone. However, ipilimumab with low dose (3 mg/kg) in untreated patients had lower ORR than

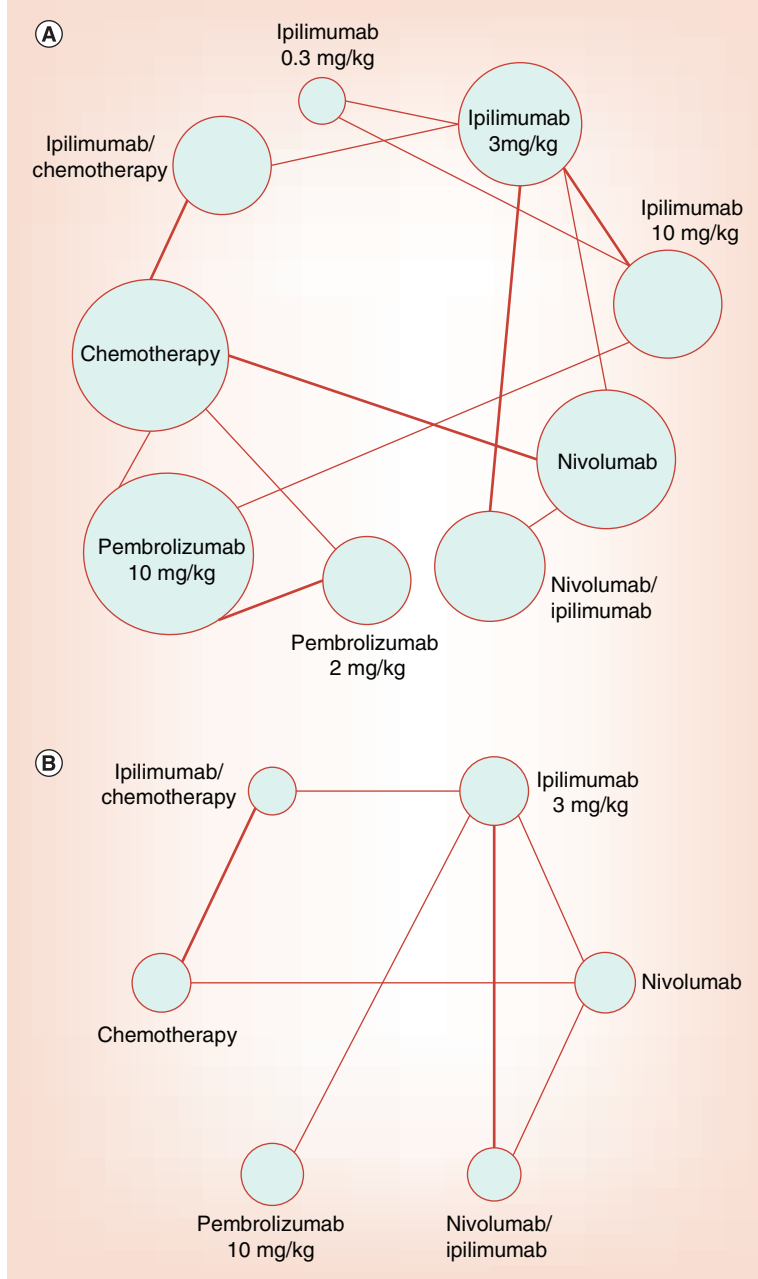

Figure 2. Evidence network of ten included studies. (A) Overall patients. **(B)** Previously untreated patients.

Table 2. Traditional meta-analysis for objective response rate of immunotherapy in advanced melanoma.

	Number of subsets	Fixed effect (95% CI)	Heterogeneity test (p-value)	Random effect (95% CI)	Egger's Test
Immunotherapy	22	0.318 (0.301, 0.336)	<0.001	0.224 (0.169,0.290)	0.011
– Without chemotherapy	19	0.329 (0.311, 0.347)	<0.001	0.222 (0.165,0.292)	0.009
– With chemotherapy	3	0.205 (0.162, 0.256)	<0.001	0.239 (0.087, 0.508)	0.729
Untreated patients	12	0.349 (0.329, 0.369)	<0.001	0.279 (0.195, 0.383)	0.256
Pretreated patients	7	0.215 (0.184, 0.249)	0.001	0.176 (0.120, 0.251)	0.017
– Previous immunotherapy	4	0.239 (0.204, 0.277)	0.541	0.239 (0.204, 0.277)	0.600
– Previous chemotherapy	3	0.075 (0.043, 0.130)	0.064	0.051 (0.015, 0.159)	0.174
Chemotherapy alone	5	0.112 (0.090, 0.139)	0.018	0.108 (0.071, 0.161)	0.778
– Untreated patients	3	0.128 (0.101, 0.160)	0.200	0.131 (0.095, 0.177)	0.493
– Previous immunotherapy	1	0.045 (0.023, 0.0870)	1.000	0.045 (0.023, 0.087)	–

chemotherapy alone. In comparison between patients treated with previous chemotherapy and immunotherapy, the ORR of immunotherapy in the group previously treated with chemotherapy (ORR: 0.051, 95% CI: 0.015–0.159) was significantly ($p < 0.001$) lower than that (ORR: 0.239, 95% CI: 0.204–0.277) in the group previously treated with immunotherapy (Table 2). Interestingly, in patients previously treated with immunotherapy, ORR of chemotherapy alone was 0.045 (95% CI: 0.023–0.087). According to our results, immunotherapy in untreated patients might be superior compared with chemotherapy alone. In addition, previous chemotherapy before immunotherapy may decrease the response of immunotherapy. However, in the meta-regression test of the current traditional meta-analysis, there was no significant difference between the ORR of immunotherapy and that of chemotherapy (0.279, 95% CI: 0.195–0.383 versus 0.131, 95% CI: 0.095–0.177, $p = 0.091$) in untreated patients. Whether the appropriate first-line treatment is immunotherapy or chemotherapy for advanced melanomas is not fully elucidated from the results of traditional meta-analysis.

Previous RCTs have investigated the therapeutic effects according to the doses of drug used [12,14,17,23]. In immunotherapy with ipilimumab, patients with high dose showed higher ORR than patients with low dose. The ORRs of high and low doses of pembrolizumab are controversial. However, in the present network meta-analysis, ORRs of pembrolizumab 10 and 2 mg/kg were higher than that of chemotherapy alone. The OR between pembrolizumab 10 and 2 mg/kg was 1.17 (95% CI: 0.60–2.19). Interestingly, ORR of ipilimumab 0.3 mg/kg was lower than that of chemotherapy alone. However, the ORRs of ipilimumab 10 and 3 mg/kg were significantly higher than that of ipilmumab 0.3 mg/kg.

Although, various targeted therapies using BRAF inhibitor have been used in advanced melanomas with *BRAF* mutation [8,26], other treatment modality for *BRAF* wild-type melanomas is needed. It has been reported that there is no significant difference in ORR between nivolumab/ipilimumab and ipilimumab in untreated melanomas patients with and without *BRAF* mutation [16]. In addition, *BRAF* mutation does not affect the ORR of nivolumab immunotherapy [22]. When melanomas shows refractory against BRAF inhibitors, recommendation of the second-line therapy is unclear. In the present study, subgroup analysis could not be performed for patients treated with BRAF inhibitors due to limited information of eligible studies. However, according to the overall results, immunotherapy could be useful for treating advanced melanomas regardless of *BRAF* mutation.

The current study has a number of limitations that should be mentioned. First, in the current study, we performed a subgroup analysis according to previous therapy. However, some eligible studies were not clearly defined for enrolled patients having previous therapy. In addition, a network meta-analysis for pretreated patients could not be performed due to limited information. Because of this limitation, cumulative studies according to previous therapy status of patients are needed. Second, the present study could not perform the analysis for the duration of response with treatment. Additional investigation on this will be useful for confirming the effect of immunotherapy.

Conclusion

In conclusion, our data showed that various immunotherapies had more effects than chemotherapy

(A)

Nivolumab/ ipilimumab								
1.58 (0.66–4.13)	Pembrolizumab 10 mg/kg							
1.84 (0.65–5.42)	1.17 (0.60–2.19)	Pembrolizumab 2 mg/kg						
1.95 (1.00–4.38)	1.23 (0.55–2.95)	1.07 (0.42–2.91)	Nivolumab					
2.94 (0.78–11.39)	1.85 (0.48–7.32)	1.58 (0.38–7.06)	1.50 (0.39–5.59)	Ipilimumab 10 mg/kg				
3.89 (1.32–10.48)	2.46 (0.85–6.20)	2.11 (0.68–5.88)	1.99 (0.74–4.43)	1.33 (0.28–5.71)	Ipilimumab/ chemotherapy			
6.71 (3.85–14.52)	4.24 (2.21–9.43)	3.65 (1.60–9.89)	3.44 (1.83–7.03)	2.30 (0.75–7.53)	1.74 (0.75–5.02)	Ipilimumab 3 mg/kg		
8.54 (3.78–21.70)	5.39 (2.52–12.18)	4.64 (2.00–11.65)	4.35 (2.33–8.18)	2.93 (0.75–11.79)	2.20 (1.20–4.83)	1.27 (0.57–2.60)	Chemotherapy	
248.69 (8.76–1453065.97)	157.43 (5.41–914076.78)	135.30 (4.44–813669.65)	126.77 (4.41–725689.40)	81.30 (3.38–481927.71)	66.14 (2.08–336800.00)	36.10 (1.35–202470.14)	29.50 (0.97–157803.38)	Ipilimumab 0.3 mg/kg

(B)

Nivolumab/ ipilimumab					
1.90 (0.43–9.72)	Nivolumab				
1.99 (0.26–20.15)	1.04 (0.12–11.23)	Pembrolizumab 10 mg/kg			
3.28 (0.40–22.59)	1.72 (0.26–8.45)	1.64 (0.11–15.99)	Ipilimumab/ chemotherapy		
7.47 (1.09–55.54)	3.92 (0.82–17.78)	3.73 (0.27–41.29)	2.29 (0.75–8.71)	Chemotherapy	
7.46 (2.44–31.16)	3.92 (0.99–18.74)	3.74 (0.66–22.25)	2.31 (0.48–16.99)	1.01 (0.19–6.82)	Ipilimumab 3 mg/kg

Figure 3. League table of network meta-analysis. (A) Overall patients. **(B)** Previously untreated patients.

alone for advanced melanomas. Combined treatment with nivolumab and ipilimumab is the most effective immunotherapy modality for advanced melanomas. Cumulative studies with various populations such as patients with previous chemotherapy and those with *BRAF* mutation will be needed for further analysis.

Future perspective

As the therapeutic efficacies of newly developed single agents or combinations of other immunological approaches can be easily compared with this study, the immunotherapy for melanoma can be rapidly innovated and applicable.

Financial & competing interests disclosure

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

No writing assistance was utilized in the production of this manuscript.

Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval or have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

Executive summary

Background

- The incidence of melanoma continues to rise during the past 40 years despite most cancer types are stable or declining.
- Advanced melanoma, including metastatic and unresectable cases, accounts for 4% of all melanoma at diagnosis.
- Melanoma is treated with a combination of surgery, traditional cytotoxic chemotherapy, targeted therapies and immune-based therapies.
- To date, the most clinically advanced checkpoint inhibitors are blocking monoclonal antibodies that target CTLA-4 and PD-1 pathways.
- Ipilimumab (antibody against CTLA-4), pembrolizumab and nivolumab (antibodies against PD-1) are the three agents approved by the US FDA for treating advanced melanoma.Methods
- A network meta-analysis using a Bayesian statistical model was performed to compare objective response rate (ORR) of various immunotherapies from 12 randomized controlled studies.

Results

- The estimated ORRs of immunotherapy and chemotherapy were 0.224 (95% CI: 0.169–0.290) and 0.108 (95% CI: 0.071–0.161), respectively.
- The ORs for ORR of nivolumab (1 mg/kg)/ipilmumab (3 mg/kg), pembrolizumab 10 mg/kg, pembrolizumab 2 mg/kg and nivolumab 3 mg/kg regimens were 8.54 (95% CI: 3.78–21.70), 5.39 (95% CI: 2.52–12.18), 4.64 (95% CI: 2.00–11.65) and 4.35 (95% CI: 2.33–11.79), respectively, compared with chemotherapy alone.
- The ORs for ORRs of ipilimumab 10 mg/kg alone, combination of ililmumab (10 mg/kg) and chemotherapy, and ipilimumab 3 mg/kg alone were 2.93 (95% CI: 0.75–11.79), 2.20 (95% CI: 1.20–4.83) and 1.27 (95% CI: 0.57–2.60), respectively, which were higher than that of chemotherapy alone.
- The ORR of ipilimumab 0.3 mg/kg alone was lower than that of chemotherapy alone.

Discussion & future perspective

- The current study was the first network meta-analysis of published randomized controlled studies to determine the therapeutic effects of various immunotherapy on advanced melanomas.
- Various immunotherapies had more effects than chemotherapy alone for advanced melanomas.
- Combined treatment with nivolumab and ipilimumab is the most effective immunotherapy modality for advanced melanomas.
- With a comparison of this network meta-analysis, the newly developed immunotherapeutic agents will be rapidly innovated and applicable.

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